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A Tool for Selective Inline Quantification of Co-Eluting Proteins in Chromatography Using Spectral Analysis and Partial Least Squares Regression

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ABSTRACT: Selective quantification of co-eluting proteins in chromatography is usually performed by offline analytics. This is time-consuming and can lead to late detection of irregularities in chromatography processes. To overcome this analytical bottleneck, a methodology for selective protein quantification in multicomponent mixtures by means of spectral data and partial least squares regression was presented in two previous studies. In this paper, a powerful integration of software and chromatography hardware will be introduced that enables the applicability of this methodology for a selective inline quantification of co-eluting proteins in chromatography. A specific setup consisting of a conventional liquid chromatography system, a diode array detector, and a software interface to Matlab[®] was developed. The established tool for selective inline quantification was successfully applied for a peak deconvolution of a co-eluting ternary protein mixture consisting of lysozyme, ribonuclease A, and cytochrome *c* on SP Sepharose FF. Compared to common offline analytics based on collected fractions, no loss of information regarding the retention volumes and peak flanks was observed. A comparison between the mass balances of both analytical methods showed, that the inline quantification tool can be applied for a rapid determination of pool yields. Finally, the achieved inline peak deconvolution was successfully applied to make product purity-based real-time pooling decisions. This makes the established tool for selective inline quantification a valuable approach for inline monitoring and control of chromatographic purification steps and just in time reaction on process irregularities.

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KEYWORDS: process analytical technology; inline monitoring; chemometrics; partial least squares regression; selective protein quantification; protein analytics; bioprocess monitoring

Introduction

Liquid chromatography continues to be the main workhorse of pharmaceutical protein purification, delivering the required product purity and high yields. However, irregularities in chromatography processes due to aging of columns and lot-to-lot fluctuations of feed concentrations, resins, or buffers can lead to reduced pool purities or yields. The real-time monitoring and control of critical process parameters in chromatography can help to manage these irregularities and ensure a stable product quality, an approach in accordance with the US Food and Drug Administration's (FDA's) process analytical technology (PAT) concept (FDA, 2004).

The selective titers of the target protein and co-eluting contaminants in the column flow-through are, among others, critical process parameters in protein chromatography and crucial for making pooling decisions or controlling the protein load onto the column. Selective quantification of proteins in the flow-through is commonly realized by fractionation of the elution peak or breakthrough curve of interest and examination of the fractions using offline analytical chromatography. This is not only elaborate and conflicting the PAT concept, but can also lead to late reaction on process irregularities and thus discarding of batches.

To overcome this analytical bottleneck, various real-time or near real-time monitoring tools for the selective titers of co-eluting proteins in the column flow-through have been developed (Bracewell et al., 1998; Fahrner and Blank, 1999; Fahrner et al., 1998; Kaltenbrunner et al., 2012; Nilsson et al., 1992; Proll et al., 2004; Rathore et al., 2008a, b; 2009, 2010). Near real-time monitoring can be realized by atline analysis,

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the performance of assays in close proximity to the process stream (FDA, 2004). Atline analysis has been applied to monitor the titers of co-eluting misfolded and correctly folded proteins by fluorescence spectroscopy in order to make pooling decisions (Rathore et al., 2009). However, atline monitoring is time-consuming, can lead to infrequent or delayed data and thus late detection of production irregularities, and is consequently rather inefficient for process control (van de Merbel et al., 1996).

A less laborious method for process monitoring, allowing frequent data collection, is the application of online monitoring, the automated diversion and measurement of samples from the production stream (FDA, 2004). Online monitoring allows the use of additional chemicals and the application of established assays. Flow-injection immunoassays (Nilsson et al., 1992) and biosensors (Bracewell et al., 1998; Proll et al., 2004) have been proposed for online monitoring of chromatography processes. However, these monitoring tools only allow the titer determination of a single protein in a multicomponent mixture. Further, online HPLC and UHPLC have been applied for selective protein quantification in multicomponent mixtures (Fahrner and Blank, 1999; Fahrner et al., 1998; Kaltenbrunner et al., 2012; Rathore et al., 2008a, b; 2010) and have successfully been used to make real-time pooling decisions (Fahrner et al., 1998; Kaltenbrunner et al., 2012; Rathore et al., 2008a, b; 2010) and to control antibody loading in protein A chromatography (Fahrner and Blank, 1999). In addition to the expensive equipment, the major disadvantage of online HPLC or UHPLC is the rather long response time (usually several minutes) (van de Merbel et al., 1996). Although efforts have been made to decrease assay times (Fahrner and Blank, 1999; Fahrner et al., 1998; Kaltenbrunner et al., 2012; Rathore et al., 2008a, b; 2010), the response time may require a slowdown of the flow rate (Kaltenbrunner et al., 2012; Rathore et al., 2008a, b; 2010) and/or the application of mathematical models to predict the protein titers in the upcoming fraction (Rathore et al., 2008a, b; 2010).

An alternative methodology for selective quantification of substances in multicomponent mixtures is spectral analysis in combination with chemometrics-based, multivariate regression techniques such as Principal Component Regression or Partial Least Squares Regression (PLS). This methodology has been promoted by the FDA's PAT guidance for process monitoring and control (FDA, 2004). It is rapid, non-invasive, cost-effective and therefore very suitable for online and inline monitoring, the continuous measurement using sensors directly inserted in the production stream (FDA, 2004). Among the available multivariate regression techniques, PLS has become the most popular due to the quality of the regression models and the simplicity of its implementation (Lavine, 2000). Especially for selective quantification using spectral data, PLS is very suitable, as it can handle the collinearity in the data (Næs and Mevik, 2001).

In the past decade, several tools using spectral analysis in combination with PLS for online or inline monitoring in biotechnology have been reported (Arnold et al., 2002; Boehl,

2003; Haack et al., 2004; Jung et al., 2002; Kara et al., 2010; Navrátil et al., 2005; Tamburini et al., 2003). In upstream-processing, fluorescence, near-infrared, and mid-infrared spectra in combination with PLS have been applied for online/inline monitoring of biomass (Arnold et al., 2002; Boehl, 2003; Haack et al., 2004; Navrátil et al., 2005; Tamburini et al., 2003), substrate (Jung et al., 2002; Kara et al., 2010; Mazarevica et al., 2004; Navrátil et al., 2005; Tamburini et al., 2003), product (Boehl, 2003; Kara et al., 2010; Mazarevica et al., 2004), and metabolite concentrations (Navrátil et al., 2005; Tamburini et al., 2003). In contrast to that, the use of spectral analysis in combination with chemometrics-based, multivariate regression techniques for downstream process monitoring is rather uncommon (Rathore et al., 2011). For instance, atline monitoring of a flocculation process for removing contaminants such as cell debris, host cell proteins, and nucleic acid could be achieved using near infrared spectroscopy and PLS (Yeung et al., 2000). During an ultrafiltration-based separation of a ternary protein mixture, atline monitoring of the selective protein concentrations in the retentate and permeate was performed by fluorescence spectra in combination with PLS (Elshereef et al., 2010).

Further, several studies in downstream processing showed that spectral analysis in combination with PLS can be used for selective offline quantification of biomolecules and declared this methodology as promising for process monitoring (Capito et al., 2013; Hansen et al., 2011, 2013; Kanga et al., 2013; Zang et al., 2013). Fourier transform mid-infrared spectroscopy in combination with PLS has been used for selective offline quantification of host cell proteins in harvested cell culture fluid (Capito et al., 2013). In the area of chromatographic purification of glycosaminoglycans, near-infrared spectroscopy combined with PLS has successfully been applied to perform selective offline quantification of the product in collected fractions (Zang et al., 2013).

In two previously published studies (Hansen et al., 2011, 2013), a methodology for selective offline quantification of proteins in multicomponent mixtures using UV-visible spectra and PLS was presented. The used spectral data were in the band of 240–300 nm, where mainly the aromatic structures of phenylalanine, tyrosine, and tryptophan contribute to UV-absorption. Due to unequal number of these amino acid residues in different protein species, protein mid-UV spectra vary and allow selective protein quantification in multicomponent mixtures using PLS. The method has been shown to work for various protein mixtures and has successfully been used as analytics for high throughput chromatography experiments using binary and ternary protein mixtures. Recently, other researchers have also confirmed the applicability of mid-UV spectra in combination with PLS for selective offline protein quantification in multicomponent mixtures (Kanga et al., 2013).

Offline analytics using mid-UV spectra in combination with PLS can increase the analytical throughput of collected fractions significantly. However, the methodology does not allow a real-time detection of process irregularities in

chromatography. Consequently, no product purity-based pooling decisions are possible. In this study, a powerful integration of software and chromatography hardware was established. This integration enables the applicability of mid-UV spectra in combination with PLS for a selective inline quantification of co-eluting proteins in chromatography. A specific setup consisting of a conventional liquid chromatography system, a diode array detector (DAD), and a software interface to Matlab[®] was established. As model system, a ternary protein mixture consisting of lysozyme, ribonuclease A, and cytochrome *c* was selected. A PLS model for the ternary protein mixture was calibrated and validated using a D-optimal onion design. The calibrated PLS model and the established specific setup were afterwards applied for a selective inline quantification and peak deconvolution of the co-eluting ternary protein mixture on SP Sepharose FF. To validate the established inline quantification tool, the elution peak was furthermore analyzed by fractionation and offline analytical chromatography. Finally, the established tool for selective inline protein quantification was adapted to make product purity-based real-time pooling decisions.

Materials and Methods

Model Proteins and Buffers

As model system, a ternary protein mixture consisting of lysozyme from hen egg white (lys), ribonuclease A from bovine pancreas (rib A), and cytochrome *c* from equine heart (cyt *c*) was selected. This system was chosen due to the stability of the proteins in aqueous solutions regarding the formation of multimers and due to their advantageous isoelectric point distribution, allowing separation at neutral pH by cation exchange chromatography. All proteins were purchased from Sigma–Aldrich, St. Louis, MO.

The buffer for the preparation of calibration and validation protein samples for the PLS model and for loading in chromatography was 20 mM sodium phosphate (pH 7). Elution was performed using a high-salt buffer consisting of 20 mM sodium phosphate as well as 500 mM sodium chloride (pH 7). All buffer components were purchased from VWR International GmbH, Bruchsal, Germany. Buffers were 0.22 µm filtrated and degassed by sonification before usage.

Liquid Handling Station

The generation of protein solutions for the calibration and validation of the PLS model were carried out using a Tecan Freedom Evo[®] 75 liquid handling station (LHS), equipped with a Liquid Handling Arm and a Robotic Moving Arm (Tecan, Männedorf, Swiss). The LHS was operated with EVOware[®] 2.3. The import of values for pipetting volumes was handled using Matlab[®] (MathWorks, Natick, MA).

Chromatographic Instrumentation

The protein spectra measurement of the calibration and validation samples as well as the chromatographic separation

were performed using a specific experimental setup consisting of a conventional liquid chromatography system, a DAD, and a software interface to Matlab[®]. The liquid chromatography system was an Akta[®] purifier 10 equipped with pump P-900, UV monitor UV-900 (10 mm optical path length), conductivity monitor C-900, pH monitor pH-900, auto-sampler A-905, and fraction collector Frac-950 (all GE Healthcare, Chalfont St. Giles, UK). The instrumentation was controlled with UNICORN[®] 5.31 software. In order to perform inline protein spectra measurements, a Dionex UltiMate[®] 3000 DAD equipped with a semi-preparative flow cell (0.4 mm optical path length) and operated with Chromeleon[®] 6.80 software (Thermo Fisher Scientific, Waltham, MA) was connected to the Akta[®] Purifier 10. The data acquisition of the DAD was triggered after a sample injection as follows: The fact of sample injection was first submitted to Matlab[®] by sending a digital signal to the Matlab[®] Data Acquisition (DAQ) toolbox using the remote connector of the Akta[®] P-900 pump in combination with an USB-6008 DAQ device (National Instruments, Austin, TX) (cf. Fig. 1). Afterwards, the signal was forwarded to an Excel Visual Basic for Application (VBA) Macro (Microsoft, Redmond, WA) using the Matlab[®] Spreadsheet Link EX toolbox. The VBA Macro then instructed the Chromeleon[®] software to start raw data acquisition using the Chromeleon[®] Software Developer Kit (Thermo Fisher Scientific). At the end of an experiment, the DAD was triggered to stop data acquisition using the same interface.

PLS Model Calibration and Validation

The calibration of a PLS model was based on a four-level D-optimal onion design, generated with MODDE (Umetrics, Umeå, Sweden). The design consisted of 29 samples,

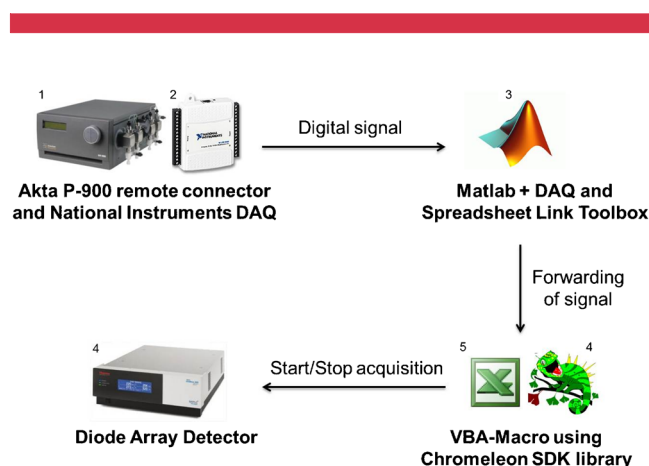


Figure 1. Trigger for the data acquisition of the DAD: the fact of sample injection or the end of an experiment was first submitted to Matlab[®] by sending a digital signal to the Matlab[®] DAQ toolbox using the remote connector of the Akta[®] P-900 pump in combination with an USB-6008 DAQ device. Afterwards, the signal was forwarded to a VBA Macro using the Matlab[®] Spreadsheet Link EX toolbox. The VBA Macro then instructed the Chromeleon[®] software to start or stop data acquisition using the Chromeleon[®] Software Developer Kit. Graphics taken from/copyright by 1: GE Healthcare, 2: National Instruments, 3: Mathworks, 4: Dionex, 5: Microsoft.

including 3 center points, and covered 19 mixing ratios and 9 concentration levels from 0 to 0.7 g/L. An additional concentration level at 0.05 g/L was added to the design as pure samples to support a good model quality at low protein concentrations. The validation of the model was performed using a three-level D-optimal onion design, generated with MODDE. The validation design consisted of 25 samples, including 3 center points, and was covering 16 mixing ratios and 8 concentration levels from 0 to 0.7 g/L. As for the calibration design, an additional concentration level with low protein concentrations of 0.06 g/L was added as pure samples. The mixing ratios of all three proteins for both the calibration and the validation sample set are shown in Figure 2A. Further, the absolute concentrations of lys and cyt *c* for both the calibration and the validation are shown in Figure 2B. The mixing ratios were prepared in AB gene polypropylene 96 deep well plates (volumetric capacity of 1 mL, Thermo Fisher Scientific) using the LHS. All mixing ratios were prepared from stock solutions, containing 3.5 g/L protein in 20 mM sodium phosphate buffer. In order to avoid systematic errors, the pipetting order of the calibration and validation samples was randomized using Matlab[®]. The spectral data measurement of the samples was realized by injecting them into the described chromatographic setup with DAD until a plateau of the UV absorption at 280 nm was reached. For each injected calibration solution, the absorption spectrum in the band of 240–300 nm with 1 nm resolution was measured. The PLS model calibration based on the obtained spectral data was finally performed with Matlab[®] using the *plsregress* function. The *plsregress* function preprocesses the spectral data by mean centering and uses the SIMPLS-algorithm for the PLS model generation.

PLS aims to reduce data by only applying significant variance in data for correlating several input variables with variables of interest. To achieve a data reduction, variables containing similar information are summarized in so-called latent variables (LVs). Thus, the number of LVs expresses a

measure of data reduction, where a low number of LVs corresponds to a high reduction. The information content included in each LV is high for the first LV and decreases for the following LVs. This means that the variance in the data can be explained by only a few LVs, leaving out higher LVs that mainly contain detector noise. The higher the number of LVs used in a PLS model, the more precise is the established model for the calibration sample set. However, the use of too many LVs can result in overfitting. In this case, the prediction of the variables of interest for other samples, such as the validation samples, decreases. Therefore, the thorough examination of the number of LVs used in a PLS model is crucial for its prediction precision (Eriksson et al., 2006). The determination of a reasonable number of LVs was realized by calculating the root mean square error (RMSE) for both calibration and validation in dependence on the numbers of LVs used in the PLS model. The final PLS model was applied to calculate the selective protein concentrations of the validation samples from their measured protein spectra. The residuals between predicted and actual protein concentrations were calculated to validate the PLS model. Normally distributed residuals below 0.05 g/L were declared as acceptable.

Inline Quantification of Co-Eluting Proteins

A HiTrap[®] 1 mL column prepacked with SP Sepharose FF (GE Healthcare) was first equilibrated with six column volumes (CV) of loading buffer and then loaded with 0.2 mg of each protein. After the protein load, the DAD was triggered to start data acquisition and recorded protein spectra in the band of 240–300 nm with 1 nm resolution. As shown in Figure 3, the spectra were transferred to Matlab[®] for selective protein quantification using the calibrated PLS model. The interface between Chromeleon[®] and Matlab[®] was realized with a VBA Macro in combination with the Chromeleon[®] Software Developer Kit. After protein loading, a wash of one

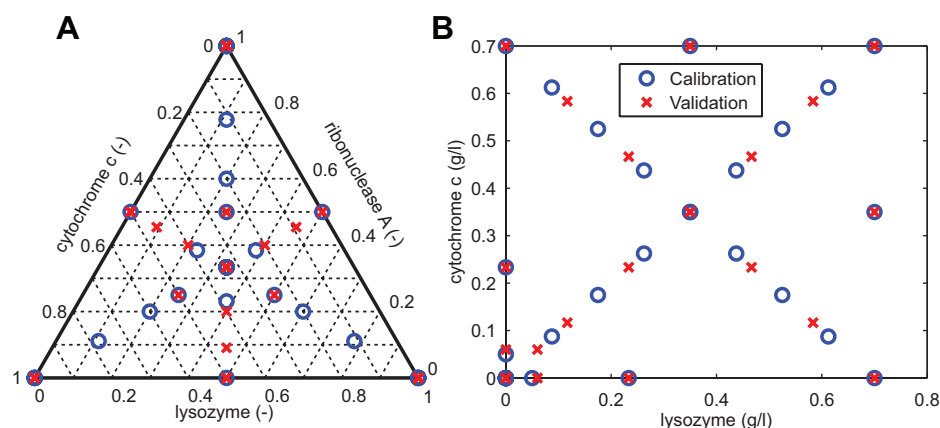


Figure 2. Mixing ratios of all three proteins (A) and absolute concentrations of cytochrome *c* and lysozyme (B) applied in the calibration and validation sample set. The calibration of the PLS model was based on a four-level D-optimal onion design, while the validation of the model was based on a three-level D-optimal onion design.

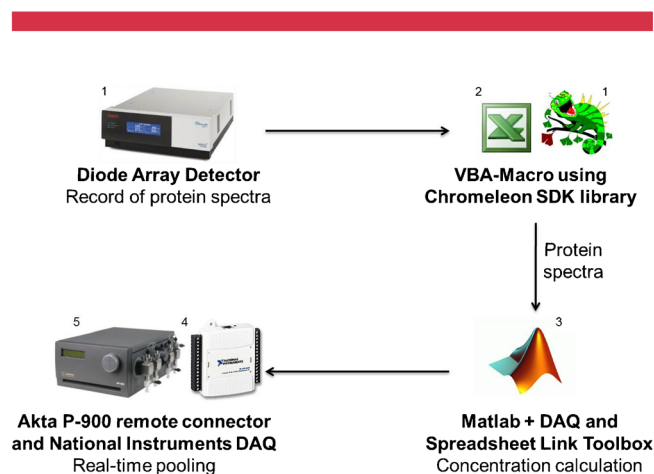


Figure 3. Selective inline protein quantification and real-time pooling decision: the DAD recorded protein spectra in the band of 240–300 nm with 1 nm resolution. The spectra were transferred to Matlab[®] for selective protein quantification using the calibrated PLS model. The interface between Chromeleon[®] and Matlab[®] was realized with a VBA Macro in combination with the Chromeleon[®] Software Developer Kit. When the criterion to start or stop pooling was detected by the Matlab[®], a digital signal was sent to the remote connector of the Akta[®] P-900 pump via the Matlab[®] Data Acquisition toolbox and the USB-6008 data acquisition device. Graphics taken from/copyright by 1: Dionex, 2: Microsoft, 3: Mathworks, 4: National Instruments, 5: GE Healthcare.

CV loading buffer was performed and the proteins were then eluted by a linear gradient from 0 to 500 mM sodium chloride in three CVs. The flow rate was 0.3 mL/min for all steps. The elution peaks were collected in a microtiter plate (Greiner BioOne, Kremsmünster, Austria) in 150 µL fractions for analytical chromatography. Based on the resulting chromatogram of the inline measurement, mass balances for all three proteins were calculated using the *trapz* function of Matlab[®].

Analytical Chromatography

In addition to selective inline protein quantification by spectral analysis, offline analytical chromatography was performed with the collected fractions, using a Dionex UltiMate[®] 3000 liquid chromatography system (Thermo Fisher Scientific). The system was composed of a HPG-3400RS pump, a WPS-3000TFC-analytical autosampler, a TCC-3000RS column thermostat, and a DAD3000RS detector. For each fraction, a Proteomix SC X-NP1.7 2.5 mL column (Sepax Technologies, Newark, DE) was first equilibrated with two CVs of loading buffer, loaded with 20 µL sample, and eluted by a linear gradient from 0 to 500 mM sodium chloride in four CVs. The flow rate was 0.5 mL/min. As for the inline quantification, mass balances were calculated by Matlab[®] based on the determined concentrations in the fractions.

Real-Time Pooling Decisions

The established tool for selective inline quantification of co-eluting proteins was applied for product purity-based

real-time pooling decisions. Therefore, rib A was declared as target protein for a first experiment and was separated from the two other proteins on the HiTrap[®] column prepacked with SP Sepharose FF with a gradient length of three CVs. In a second experiment, *cyt c* was selected as target and separated from the two other proteins with a gradient length of five CVs. The flow rate was 0.3 mL/min for both experiments. The criterion to start pooling was a target protein concentration above 0.002 g/L as well as a mass fraction of the target above 0.8 in the column flow-through. As soon as this criterion was detected by the Matlab[®] function that was calculating the selective protein concentrations, a digital signal was sent to the REMOTE connector of the Akta[®] P-900 pump via the Matlab[®] Data Acquisition toolbox and the USB-6008 data acquisition device (cf. Fig. 3). This digital signal was then used in UNICORN[®] as criterion to start pooling. The criterion to stop pooling was a mass fraction of the target protein below 0.8. The end of pooling was managed using the interface between Matlab[®] and UNICORN[®] that was described above. The time span between the detection of the criterion to start or to stop pooling by Matlab[®] and the reaction on the signal by UNICORN[®] was below 1 s. Based on the selective inline quantification, the pool purities and yields were calculated.

Results and Discussion

Product purity-based real-time pooling decisions in liquid chromatography require a fast and reliable analysis of the column flow-through. This work demonstrates the applicability of a methodology for selective protein quantification (Hansen et al., 2011, 2013) as an inline monitoring tool in chromatography. The methodology uses a PLS model to correlate protein UV-visible spectra with selective protein concentrations. The inline application of the methodology for selective protein quantification requires a specific experimental setup consisting of a liquid chromatography system, a DAD, and a software interface to Matlab[®]. The applied detector type (DAD) was different from the detector used in (Hansen et al., 2011, 2013). Consequently, a new PLS model for the used ternary protein mixture needed to be calibrated and validated. The calibrated model and the established experimental setup were afterwards applied for selective inline protein quantification in the column flow-through. The selective inline protein quantification was thoroughly validated by comparing it with offline analytical chromatography. In order to prove the practical relevance of this new monitoring tool, it was shown to be applicable for real-time pooling decisions.

PLS Model Calibration and Validation

PLS aims to reduce data by only applying significant variance in data for correlating several input variables (here protein spectra) with variables of interest (here selective protein concentrations). To achieve a data reduction, variables containing similar information are summarized in LVs. A

thorough examination of the number of LVs used in a PLS model is crucial for its prediction precision (Eriksson et al., 2006). In order to determine the optimal number of LVs, the root mean square error for the calibration samples (RMSEC) and the root mean square error for the prediction (RMSEP) of the selective concentrations in the validation sample set were calculated for eight different PLS models with one to eight LVs used in the calibration. Figure 4A shows the calculated RMSECs and RMSEPs of the eight PLS models in dependence on the number of LVs used in the model calibration. For both lys and cyt *c*, a significant decrease in the RMSEC and RMSEP was observed when two instead of one LV was used, which shows that the second LV contained a major part of the information about the concentration of these two proteins. The first LV mainly contained information about the total protein concentration (data not shown). The major part of the information about the rib A concentration was in the third LV. While the RMSEC was found to decrease for all proteins with increasing number of LVs, the RMSEP decreased continuously for lys and cyt *c* and exhibited a minimum for ribonuclease A at six LVs. Additionally, a significant decrease of the RMSEC and RMSEP of cyt *c* was observed, when the number of LVs was increased from five to six. Therefore, a PLS model with six LVs was selected. The six LVs explained 99.96% of the variance in the protein concentrations of the calibration sample set and 99.92% of the variance in the protein concentrations of the validation sample set.

For the PLS model with six LVs, the relative frequency distribution of the residuals between predicted and actual protein concentrations of the validation sample set was

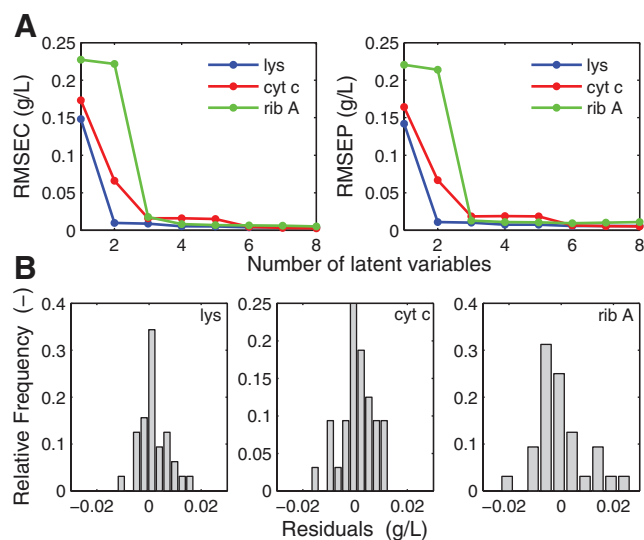


Figure 4. A: Calculated root mean square error for the calibration (RMSEC) and validation (RMSEP) with respect to the number of latent variables used for the PLS Model calibration. B: Relative frequency distribution of the residuals of lysozyme, cytochrome *c*, and ribonuclease A in the model validation for six latent variables.

calculated for all three proteins. The results are shown in Figure 4B. The residuals of all three proteins were nearly normally distributed, which indicates that systematic errors occurred during the PLS model calibration are rather unlikely. The residuals lay between -0.0121 and 0.0171 g/L for lys, -0.0168 and 0.0126 g/L for cyt *c*, and between -0.0232 and 0.0265 g/L for rib A, implying a high prediction precision of the calibrated PLS model with six LVs.

Inline Quantification of Co-Eluting Proteins

The calibrated PLS model was applied for an inline quantification of co-eluting lys, cyt *c*, and rib A during a chromatographic separation on SP Sepharose FF with a gradient length of three CVs. The established experimental setup consisting of an Akta[®] liquid chromatography system, a DAD, and a software interface to Matlab[®], allowed a selective protein quantification in real-time. Figure 5 shows the resulting chromatograms of the inline measurement. The predicted, selective protein concentrations by the PLS model (cf. Fig. 5B, solid lines) are compared with both the absorption at 280 nm (A_{280}) and at 527 nm (A_{527}) (cf. Fig. 5A) and with the determined protein concentrations in the collected fractions using offline analytical chromatography (cf. Fig. 5B, dashed lines). Good agreement was found between the retention volume of the first peak in the A_{280} signal and the rib A concentration profile from the inline measurement. The retention volume of cyt *c* from the inline measurement corresponded to the peak maximum of the A_{527} signal, which is characteristic for cyt *c*. Further, both

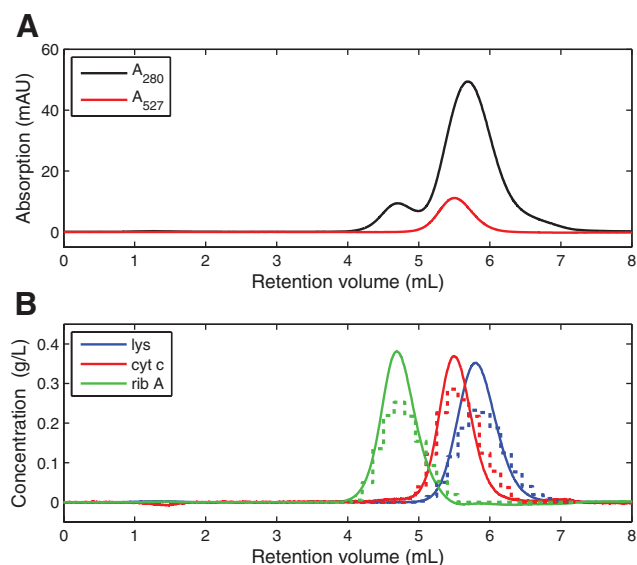


Figure 5. Resulting chromatograms of the inline measurement and offline analytical chromatography. A: Absorption at 280 and 527 nm plotted against the retention volume. B: Predicted protein concentrations by the PLS model (solid lines) and determined protein concentrations in collected fractions using offline analytical chromatography (dashed lines) plotted against the retention volume.

the selective inline quantification and the offline analytical chromatography resulted in the same retention volume for all three model proteins and exhibited good agreement in the peak flanks. However, the protein peaks determined from the offline analytical chromatography exhibited negligibly more tailing to some degree, which might be explained by additional dispersion in the capillary between DAD and fraction collector. The good agreement with both the absorption measurement and the results from the analytical chromatography prove the precision and applicability of the established inline quantification tool.

Regarding the protein concentration at the peak maximums of all three proteins, small deviations were found between both analytical methods (cf. Fig. 5A). A slightly better agreement in the protein concentrations at the peak maximums might have been achieved, if the resolution of the collected fractions had been increased. However, the inline protein quantification tended to show slightly higher protein concentrations in the peak maximums compared with analytical chromatography.

Table I shows the calculated mass balances for both analytical methods. The mass balances resulted in a range between 108% and 117% for the inline quantification and in a range of 89–108% for analytical chromatography. While the inline quantification was found to slightly overestimate the protein masses in the column flow-through, analytical chromatography resulted in mass balances below 100% for lys and rib A. The slight overestimation of protein masses by the inline quantification mainly resulted from an overestimation of the protein concentrations at the peak maximums. However, in the validation of the PLS model, an overestimation for samples with similar concentrations as found in the peak maximums was not observed. The residuals between actual and predicted protein concentrations in the validation were all below 0.03 g/L (cf. Fig. 4B). Consequently, an error in the PLS model calibration for this concentration area can be excluded. A substantiated explanation for the observed overestimation is, however, currently not at hand. But although the mass recovery of the analytical chromatography was marginally more accurate, the predicted selective masses by the PLS model are acceptable regarding the fact that they can be determined inline. Consequently, the established tool can also be applied for a rapid determination of pool yields.

Real-Time Pooling Decisions

The established tool for selective inline quantification of co-eluting proteins was finally applied for real-time pooling

Table I. Mass balances (in %) for lys, cyt *c*, and rib A for both selective inline protein quantification and offline analytical chromatography.

	Analytical chromatography	Inline quantification
Lys	95.80	116.60
Cyt <i>c</i>	108.68	113.45
Rib A	88.78	107.85

decisions and a rapid determination of pool purities and yields. In a first exemplary experiment, rib A was declared as target protein and separated from the “contaminants” lys and cyt *c* on SP Sepharose FF with a gradient length of three CVs. In a second experiment, cyt *c* was the target and attempted to be separated using a gradient length of five CVs. For both experiments, the criterion to start pooling was a target protein concentration above 0.002 g/L and a mass fraction of the target above 0.8. The criterion to stop pooling was a mass fraction of the target below 0.8. The established, selective inline quantification tool as well as the established interface between Matlab[®] and UNICORN[®] allowed an automated, product purity-based pooling in real time. The resulting chromatograms of both experiments are shown in Figure 6. The black lines in the chromatograms visualize the pool borders that were automatically detected by Matlab[®] and forwarded to the software UNICORN[®] which was operating the liquid chromatography system. Finally, the mass balances gained by the selective inline quantification and the pool borders were used to calculate the product yield and purity in the collected pools. The calculated yields were 88.8% for rib A and 36.3% for cyt *c*, respectively. The determination of the pool purity resulted in 96.9% for the rib A pool and in 83.5% for the cyt *c* pool.

Conclusion and Outlook

A tool for selective inline quantification of co-eluting proteins in chromatography was successfully established. A specific setup consisting of a conventional liquid chromatography

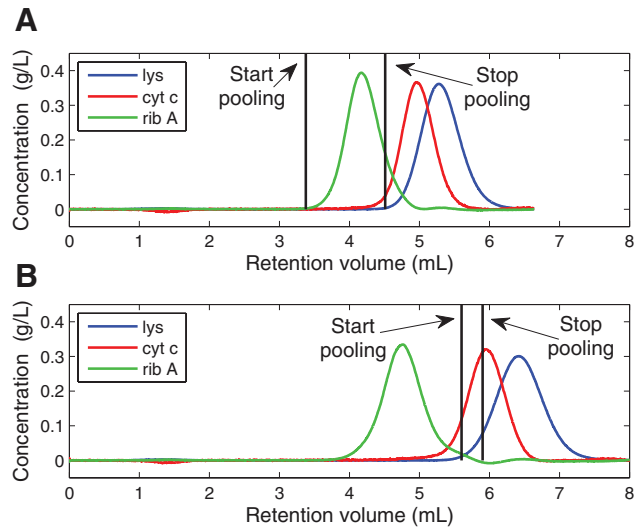


Figure 6. Resulting chromatograms of the product purity-based real-time pooling during a chromatographic separation of lys, cyt *c*, and rib A on SP Sepharose FF. The black lines in the chromatograms visualize the pool borders that were automatically detected by Matlab[®] and forwarded to the software UNICORN[®]. **A:** Rib A was declared as target protein and separated with a gradient length of three CVs. **B:** Cyt *c* was declared as target protein and separated with a gradient length of five CVs.

system and a DAD was developed to perform inline measurements of protein spectra. To correlate the protein spectra with selective protein concentrations, a PLS model was calibrated. An established software interface allowed a selective inline quantification of the co-eluting proteins by Matlab[®] using the calibrated PLS model and the protein spectra. The tool for selective inline quantification was rigorously compared with offline analytical chromatography using collected fractions and found to deliver precise results for both the peak flanks and the retention volumes of co-eluting proteins. Further, a comparison between the mass balances based on analytical chromatography and inline quantification showed that the established tool can be applied for a rapid determination of pool yields. Using the gained peak deconvolution by the inline quantification tool, precise purity-based real-time pooling decisions were possible. Consequently, a reliable inline monitoring tool was established that can be used to detect process irregularities. The presented tool has a great potential to manage irregularities in preparative chromatography processes by real-time pooling. Future work will focus on more complex applications with pharmaceutically important proteins and separation challenges. One first step in this context will be the elimination of the necessity of having each protein as a pure component for the PLS model calibration. This could be achieved by performing the calibration with samples from the chromatographic separation process to be monitored. As each component in a process needs to be accounted for in the calibration in order to use the inline quantification tool, a possible application could be the inline monitoring of co-eluting product and product-related impurities during an intermediate purification or polishing step.

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